

[ISOM2010] INTRODUCTION TO INFORMATION SYSTEMS

E-COMMERCE

S7

PROF. KIM, YONGSUK

KEY CONCEPTS IN DIGITAL MARKETS

Digital Markets vs. Traditional Markets

	DIGITAL MARKETS	TRADITIONAL MARKETS
Information asymmetry	Asymmetry reduced	Asymmetry high
Search costs	Low	High
Transaction costs	Low (sometimes virtually nothing)	High (time, travel)
Delayed gratification	High (or lower in the case of a digital good)	Lower: purchase now
Menu costs	Low	High
Dynamic pricing	Low cost, instant	High cost, delayed
Price discrimination	Low cost, instant	High cost, delayed
Market segmentation	Low cost, moderate precision	High cost, less precision
Switching costs	Higher/lower (depending on product characteristics)	High
Network effects	Strong	Weaker
Disintermediation	More possible/likely	Less possible/unlikely

Source: Laudon and Laudon (2018)

- **Types of Business in Digital Markets**

- B2B
- B2C
- C2B
- C2C

- **Revenue Models in Digital Markets**

- Advertising
- Sales
- Subscription and Freemium
- Transaction Fee
- Affiliate

- **Long-tail and Recommendation Algorithms**

- Collaborative Filtering
- Content-based Filtering

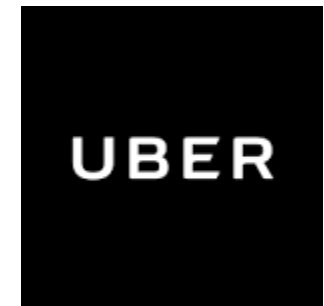
TYPES OF BUSINESS IN DIGITAL MARKETS

Types of Business



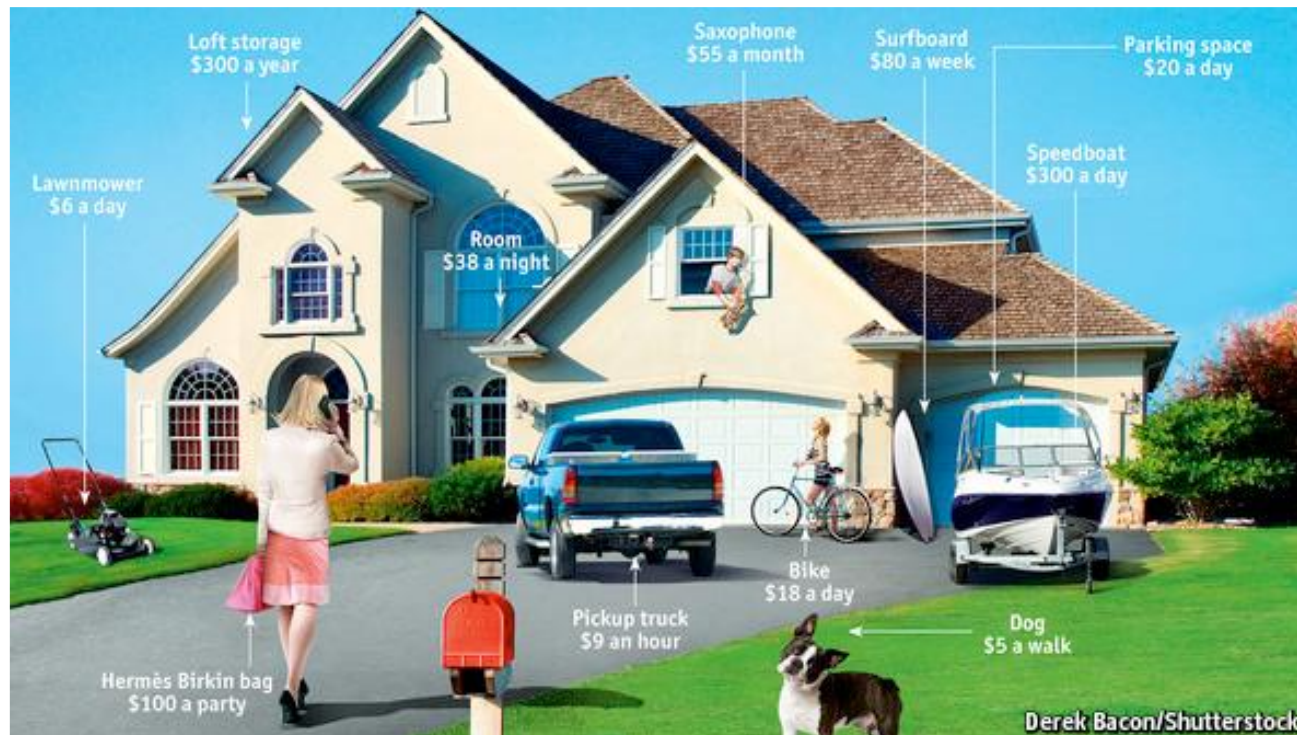
- **Business-to-consumer (B2C):** most visible
- **Business-to-business (B2B):** greatest dollar amount of transactions
- **Consumer-to-consumer (C2C):** greater geographic reach
- **Consumer-to-business (C2B):** crowdsourcing

Crowd (Sharing) economy: C2C(P2P) Model



Crowd (Sharing) economy: C2C(P2P) Model

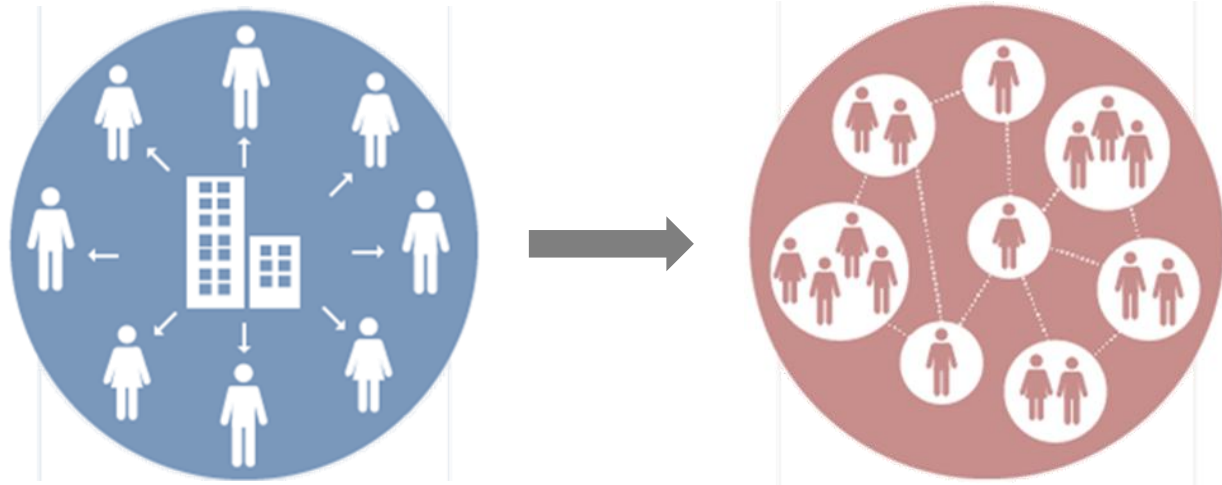
- **Extract value** from **the asset we already have**, physical or intangible, dividing them into **space and time**, in order to be **consumed as a service**
- Platforms allow pricing, meeting and sharing



Crowd (Sharing) economy: C2C(P2P) Model

● Peer to Peer Networks

- Traditional models of consumption are based on business-to-customer (B2C) transactions



- New Models of consumption are emerging and connect **spare capacity (or underutilized assets/resources)** to demand using **peer-to-peer (P2P) transactions: Consumer-to-consumer (C2C) business**
- No need for inventory
- E.g., Car ownership and car rental → Car pooling and ride sharing
- E.g., Hotel, B&B and hostels → Peer rental and house swapping

SCALABILITY OF AIRBNB



90 Years to build

697,000 Rooms

80 COUNTRIES



5 Years to build

2,000,000+ Listings

192 COUNTRIES

SCALABILITY OF AIRBNB



Brian Chesky

@bchesky



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Marriott wants to add 30,000 rooms this year. We will add that in the next 2 weeks.

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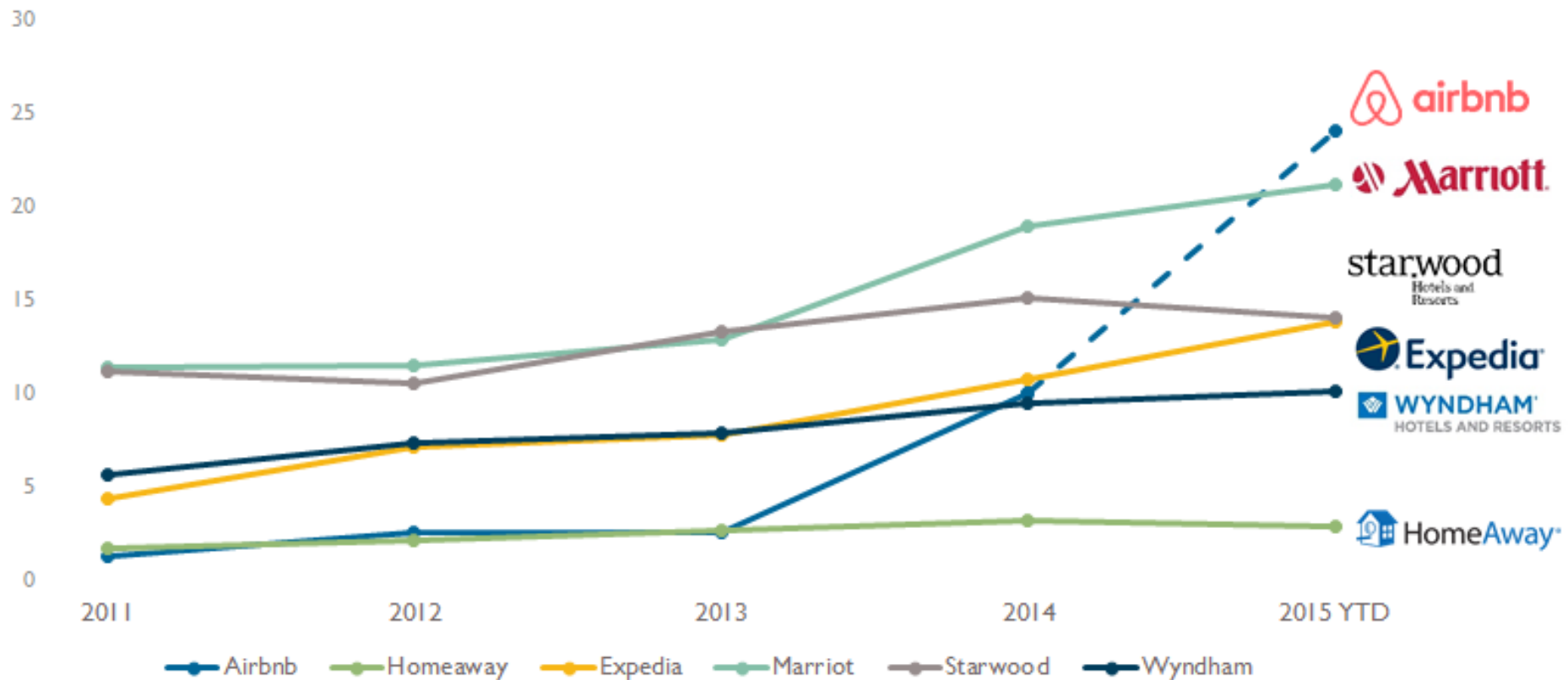
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11:11 PM - 10 Jan 2014

SCALABILITY OF AIRBNB

AirBnB vs. Public Competitors: Valuations Over Time (\$B)
2011 - 2015 YTD (6/18/2015)



*AirBnB did not raise a round between 2012 and 2014, and therefore its valuation stayed the same during this time

**Valuations were taken at dates where AirBnB raised. 2013 data was taken at 6/1/2013

TYPES OF BUSINESS IN DIGITAL MARKETS

Types of Business



- **Business-to-consumer (B2C):** most visible
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Crowdsourcing



Crowdsourcing

“The act of taking a job traditionally performed by a designated agent (usually an employee) and outsourcing it to an undefined, generally large group of people in the form of an open call.”

Jeff Howe



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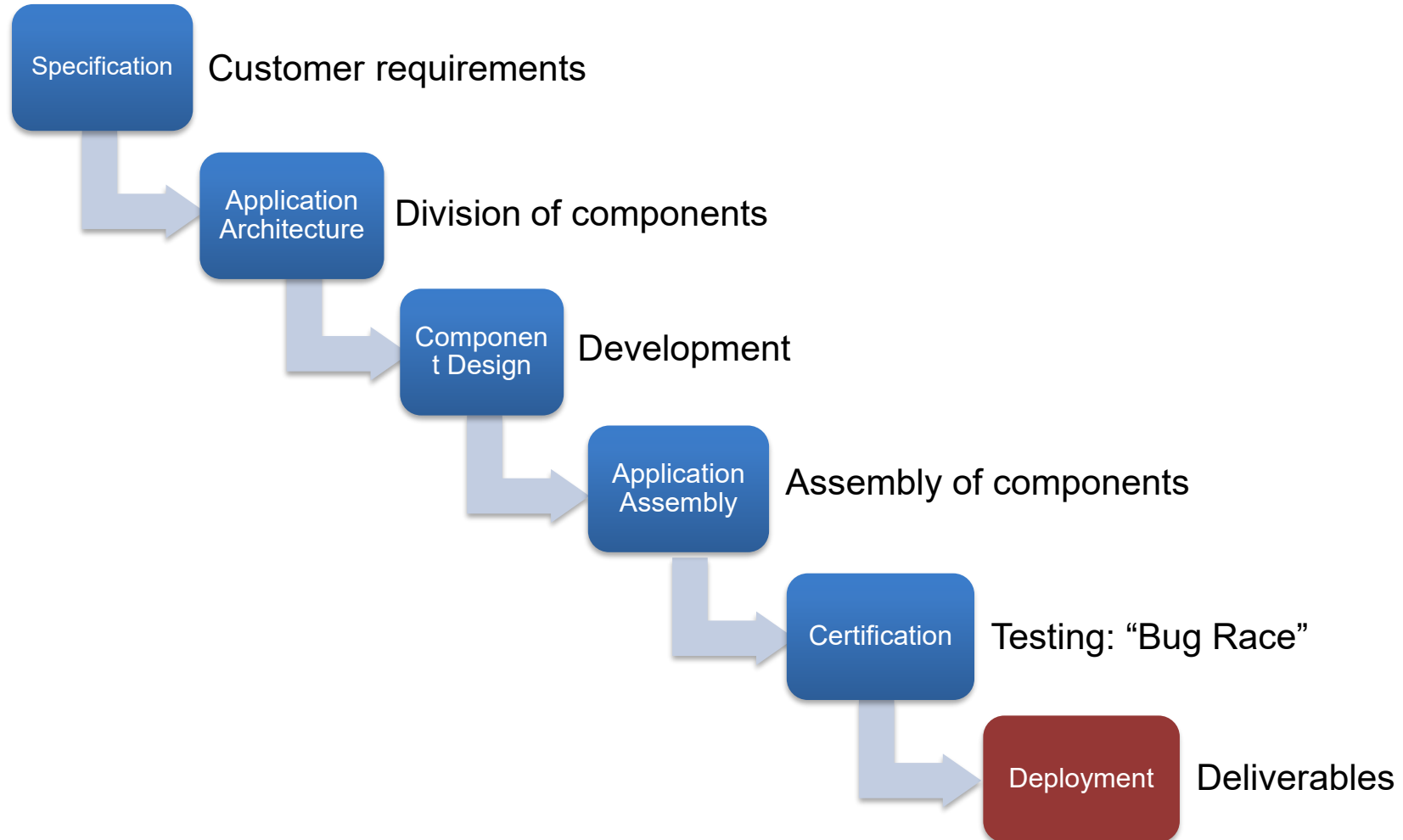


- The company uses competition (“challenges”) to build software, graphic design and others
 - Software development
 - Graphic Design
 - Data Science: Scientific Algorithm Development
- A community of 884,000+ members
- 64,225 open prizes
- Clients include Alcatel-Lucent, AOL, Cisco, Facebook, Harvard University, Microsoft, PayPal, UBS, US NSA and etc.

COMPETITION-BASED SW DEVELOPMENT



[topcoder][™]



COMMON BUSINESS MODELS IN DIGITAL MARKETS

Transaction Fee

- Charge a fee for enabling or executing a transaction
- Don't have to physically own the service or content



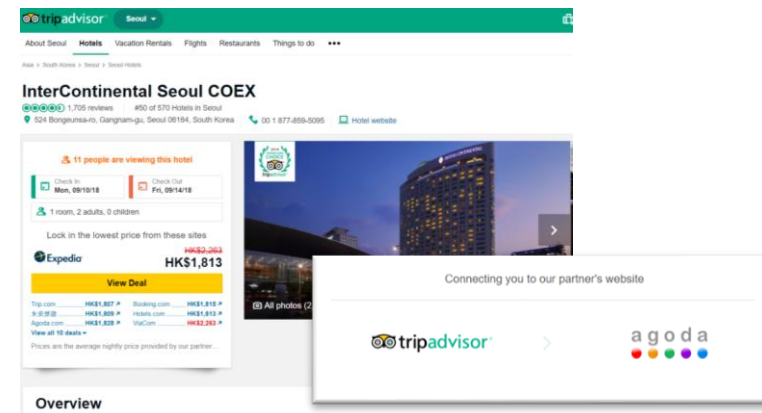
Advertising

- Charge advertisers the right to place ads on the site
- Most widely used method of generating revenue
- Higher ad fees: *attract a large number of users or keep users on site for any length of time*



Affiliate

- Receive a referral fee or percentage of sales each time you steer customers to affiliated sites



COMMON BUSINESS MODELS IN DIGITAL MARKETS

● Subscription

- Offers its users content or services and charges a subscription fee for access to some or all of its offerings.
- Provide content or services that users perceive
 - worth the cost (high-value-added)
 - not readily available elsewhere nor easily replicated

NETFLIX



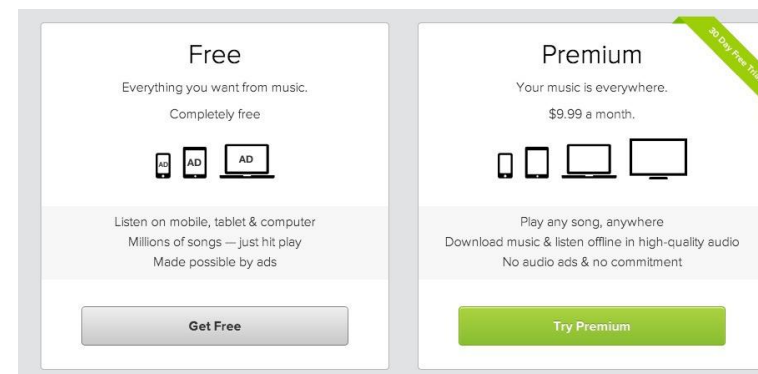
● Sales

- Sell products, information, or services directly to users



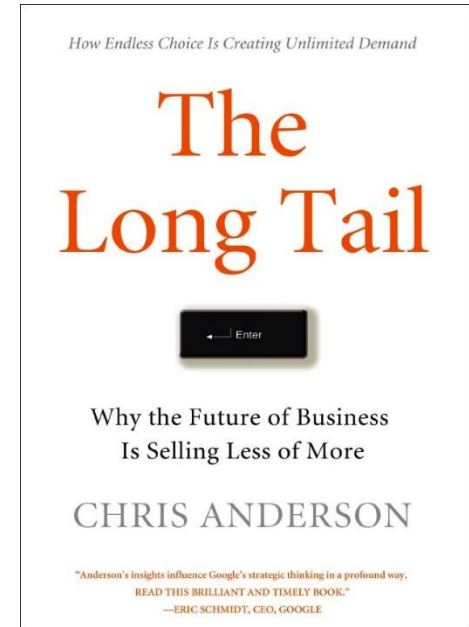
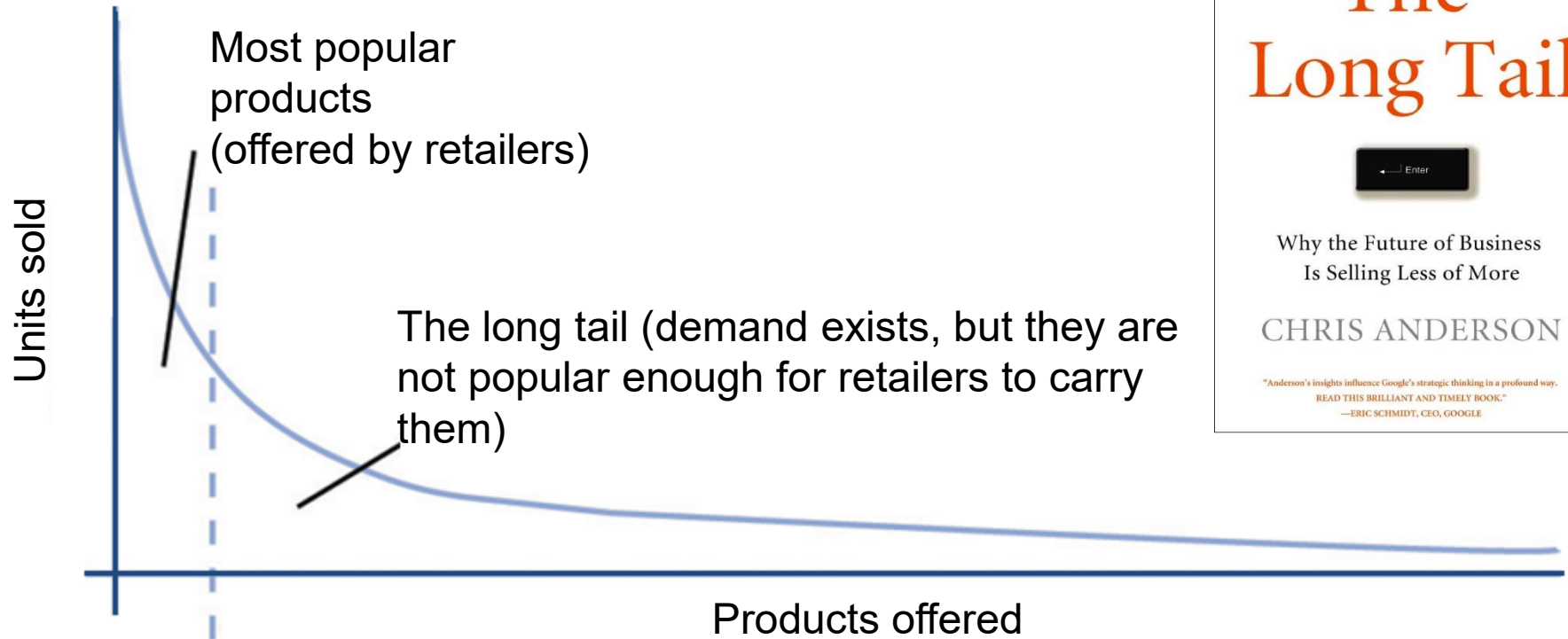
** Free/Freemium

- Provide basic content or services free but charge a premium for special features
- Entice users to your site with freebies and then convert them to paying customers



THE LONG-TAIL IN THE MARKET

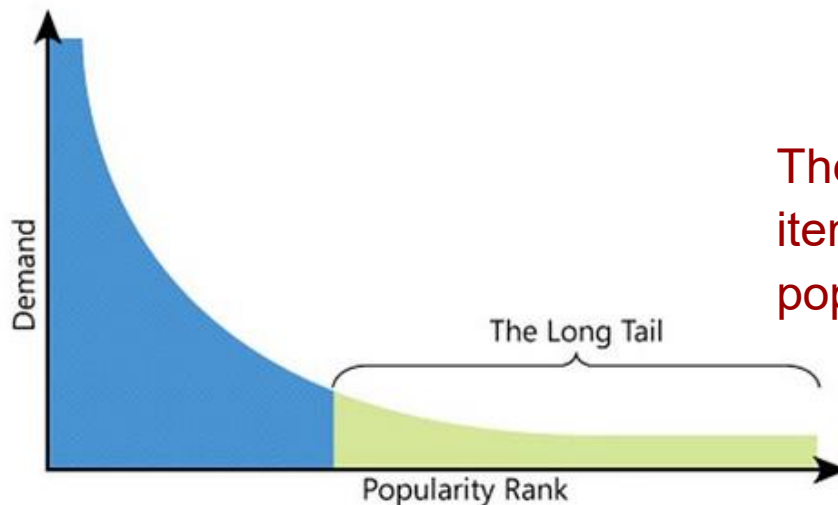
📌 *Long-Tail Theory*



PARETO PRINCIPLE VS. LONG TAIL THEORY

▣ *Pareto Principle vs. Long Tail Theory*

- **Pareto Principle:** 80% sales comes from 20% of products, selling fewer popular items in large quantities (the “head” or “superstars”)
- **Long Tail Theory:** selling a large number of unique items (hard-to-find items) with relatively small quantities sold of each (the “long tail”)



The total volume of low popularity items exceeds the volume of high popularity items.

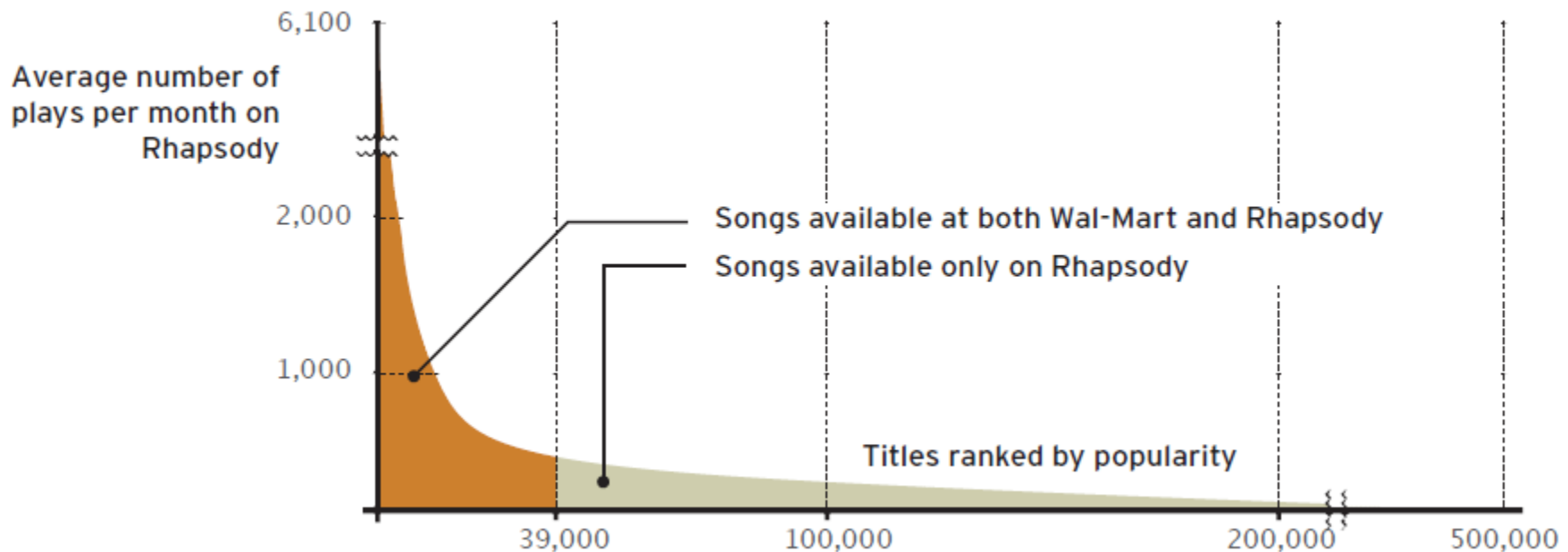
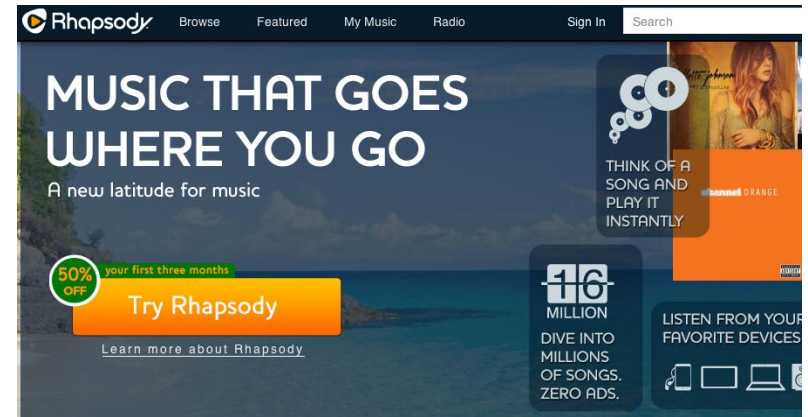
The Pareto Principle is replaced by the Long Tail Theory in Digital Markets!

THE LONG-TAIL IN THE MARKET

Long-Tail Theory



vs



Source: <http://changethis.com/manifesto/10.LongTail/pdf/10.LongTail.pdf>

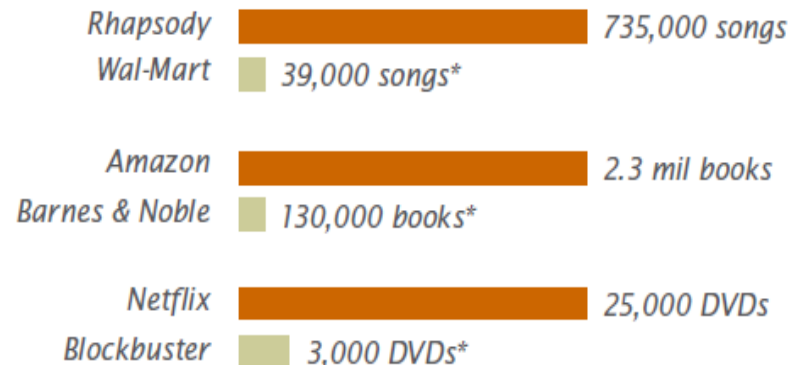
IT-DRIVEN LONG-TAIL CONSUMPTION

- IT helps form millions of *niche markets*.
- *Near-zero-cost dissemination of information* about products
- IT enables *easy searching for that information*.
 - Tools such as search engines, *recommendation* software, and sampling tools allow customers to find niche products.

	DIGITAL MARKETS	TRADITIONAL MARKETS
Information asymmetry	Asymmetry reduced	Asymmetry high
Search costs	Low	High
Transaction costs	Low (sometimes virtually nothing)	High (time, travel)

TOTAL INVENTORY

* inventory in a typical store



RECOMMENDATION SYSTEMS HELP FIND LONG-TAIL ITEMS

Yesterday, I searched for the hoodie

Shop by
Department ▾

Search

Go

Hello, Dongwon
Your Account ▾

Your
Prime ▾

 **Cart** ▾

Wish List ▾

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[Sports & Outdoors](#) > [Clothing](#) > [Men](#) > [Hoodies](#)



Roll over image to zoom in

Under Armour Men's Armour® Fleece Team Hoodie

by [Under Armour](#)

★★★★★ ▾ 48 customer reviews

Price: ~~\$36.99~~ - \$60.88

Color: **Black/White**



Size:

Select Size ▾

- Full, loose fit for enhanced range of motion & breathable comfort no matter where your workout takes you
- Armour Fleece fabrication has a brushed inner layer and a quick-drying, smooth outer layer
- Soft inner layer traps heat to keep you warm and comfortable
- Signature Moisture Transport System wicks sweat away from the body
- Three-piece hood construction for superior fit

Share    

To buy, select **Size**
Choose from options to the left



Add to Cart

Add to Wish List

RECOMMENDATION SYSTEMS HELP FIND LONG-TAIL ITEMS

My Amazon main page today

The screenshot shows the Amazon homepage with the following elements:

- Header:** Amazon Prime logo, navigation links (Dongwon's Amazon.com, Today's Deals, Gift Cards, Sell, Help), and a search bar.
- Left Sidebar:** A list of categories to shop by, including Unlimited Instant Videos, MP3s & Cloud Player, Amazon Cloud Drive, Appstore for Android, Kindle E-readers & Books, Kindle Fire Tablets, Amazon Fire TV, Books & Audible, Movies, Music & Games, Electronics & Computers, Home, Garden & Tools, Beauty, Health & Grocery, Toys, Kids & Baby, Clothing, Shoes & Jewelry, Sports & Outdoors, Automotive & Industrial, and Full Store Directory.
- Main Banner:** Amazon Fashion ULTIMATE HOLIDAY Gift Guide, featuring various clothing items and the text "IRRESISTIBLE SURPRISES FOR EVERYONE ON YOUR LIST".
- Cyber Monday Deals:** A row of circular icons for Cyber Monday Deals Week, Deal of the Day, Electronics, Toys & Games, and Amazon Devices.
- Related to Items You've Viewed:** A section with two columns of product recommendations. The first column is labeled "You viewed" and the second is labeled "Customers who viewed this also viewed".

You viewed	Customers who viewed this also viewed					
						
Under Armour Men's Charged Cotton... ★★★★★ (10) \$37.99 - \$73.18	Under Armour Men's Armour® Fleece... ★★★★★ (99) \$36.00 - \$69.99	Under Armour Men's Armour® Fleece... ★★★★★ (46) \$36.99 - \$60.88	Under Armour Men's Charged Cotton... ★★★★★ (6) \$38.99 - \$64.99	Under Armour Men's UA Tech™ Fleece... ★★★★★ (52) \$38.74 - \$59.99	Under Armour Men's Charged Cotton... ★★★★★ (51) \$34.99 - \$49.88	Under Armour Men's UA Charg Cotton... ★★★★★ (2) \$35.99

[View or edit your browsing history](#)

- Related items you have viewed shows:
 - What you have viewed
 - Customer who viewed this also viewed

RECOMMENDATION SYSTEMS HELP FIND LONG-TAIL ITEMS

*Amazon and Netflix generates **X** % of their sales through the recommendation lists*

$$30 < X < 70$$



📉 Leveraging Data Asset

● **Definition** (Xiao and Benbasat 2007)

“software agents that elicit the interests or preferences of individual customers for products and make recommendations accordingly”

- **Retailer could tailor recommendations based on availability of products and individual taste**

- Identify and track users
- Track user activity
- Store user activity
- Generate user models
- Serve recommendations

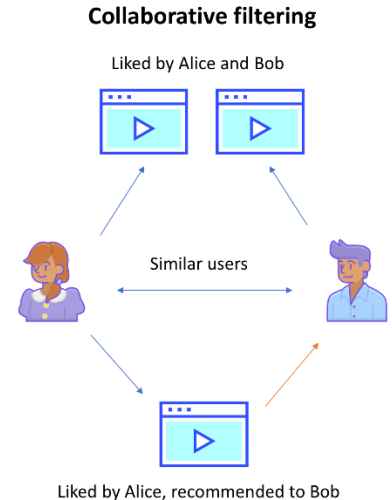


RECOMMENDATION SYSTEMS

▣ *Types of Recommendation Algorithms*

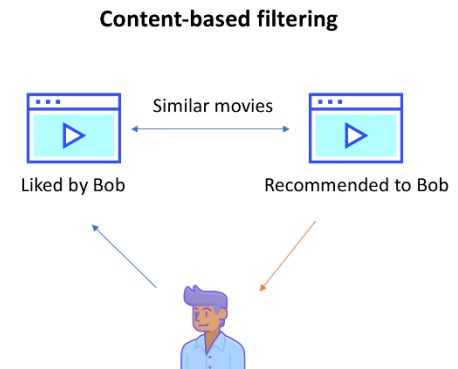
● Collaborative Filtering (CF)

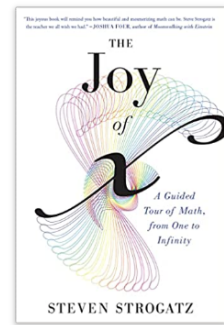
- Collecting and analyzing a large amount of information on users' behaviors, activities or preferences and *predicting what users will like based on their similarity to other users*
 - “People who viewed this also viewed”
- Need large amount of information
- The most prominent approach to generate recommendations



● Contents based Filtering

- Recommend items that *similar to those that a user liked in the past*
- Predicting what users will like based on the similarity between items (features of items)
 - “Based on your previous purchase history”
- Need little information to get started but limited in scope





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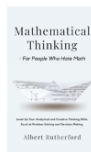
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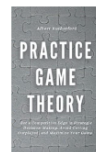
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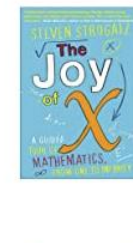


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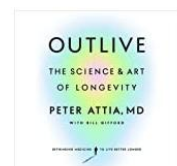


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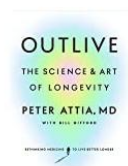


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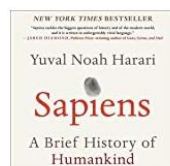
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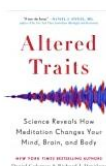
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